

Maharaja Surajmal Institute of Technology



Electricity Consumption Analysis and Prediction of New Delhi

Using Data Analytics and Machine Learning

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Dataset: 12 Months Data Of Delhi Electricity Board 2022

Abstract

This project analyzes electricity consumption patterns for New Delhi using daily time-slot records from the Delhi Electricity Board 2022 dataset. The work combines data cleaning, exploratory data analysis, visualization, and machine learning to identify demand trends across months, days, hours, and distribution companies. The cleaned dataset contains high-frequency dispatch records with measurements for Delhi and major distribution areas, and the analysis notebook derives a Total Consumption field from BRPL, BYPL, NDPL, NDMC, and MES readings. A Random Forest Regressor is then used to predict consumption from calendar and time-based features. The results show clear seasonal and intraday demand variation, with June recording the highest aggregate consumption among the available records and the afternoon period around 15:00 emerging as the highest average demand hour.

Keywords

Electricity consumption, New Delhi, data analytics, exploratory data analysis, Random Forest Regressor, demand prediction, time-series features.

Project Objectives

- Collect and merge daily electricity CSV files into a single analytical dataset.
- Clean the dataset by checking missing values, duplicate rows, and date-time consistency.
- Analyze daily, monthly, hourly, weekday, and distribution of company-wise consumption patterns.
- Identify peak consumption periods and important operational insights.
- Train and evaluate a machine learning model for electricity consumption prediction.
- Present the final results in a formal Word report with tables, charts, and page numbers.

Scope of Work

The report focuses on the available 2022 New Delhi electricity dispatch files in the project workspace. The original Kaggle dataset is described as a Delhi Electricity Board 2022 record; the local files used for computation cover the available date range from 2022-01-01 to 2022-12-10.

Dataset Description

Data Source: Kaggle - Delhi Electricity Board Record - 2022. The dataset link used for this project is https://www.kaggle.com/datasets/adityasartape/12-months-data-of-delhi-electricity-board-2022/data?select=delhi_dispatch.

According to the dataset description, the records were collected using web scraping techniques with Python's BeautifulSoup library. The dataset is useful for studying monthly usage, peak demand behavior, seasonal variation, energy sector planning, data science experiments, and climate impact studies.

Important Dataset Points

- Daily files follow the naming format `delhi_DD_MM_YYYY.csv`, which allows the date to be extracted directly from the file name.
- The TIMESLOT column records intraday intervals, enabling hourly and peak-load analysis.
- The company columns BRPL, BYPL, NDPL, NDMC, and MES represent electricity distribution areas used to compute Total_Consumption.

- The DELHI column is retained as the overall Delhi-level reading and is highly correlated with the computed total.
- The cleaned dataset contains no missing values or duplicate rows after preprocessing.

Dataset Attribute	Value
Raw daily CSV files	344
Cleaned records	92,268
Cleaned columns	16
Unique dates	337
Available date range	2022-01-01 to 2022-12-10
Missing values after cleaning	0
Duplicate rows after cleaning	0
Average intervals per day	273.79

Column	Description
TIMESLOT	Time interval of the record, generally at five-minute frequency.
DELHI	Overall Delhi electricity reading in the source file.
BRPL	BSES Rajdhani Power Ltd. distribution area reading.
BYPL	BSES Yamuna Power Ltd. distribution area reading.
NDPL	North Delhi Power Ltd. distribution area reading.
NDMC	New Delhi Municipal Council area reading.
MES	Military Engineer Services area reading.
Total / Total_Consumption	Aggregate consumption field used for analytics and prediction.

Methodology

The project workflow follows the three notebooks in the workspace: data_cleaning.ipynb, data_analysis.ipynb, and prediction_model.ipynb. The same logic is used here to generate the report charts and model results.

Step	Work Performed
1. Data Collection	Read all daily CSV files from month-wise folders under Data/.
2. Date Extraction	Extract date from each file name using the delhi_DD_MM_YYYY.csv format.
3. Data Merging	Concatenate all daily files into one dataset for analysis.
4. Cleaning	Check missing values, remove duplicates, and standardize TIMESLOT as text.
5. Feature Engineering	Create Year, Month, Day, Day_Name, DayOfWeek, Hour, and Total_Consumption.
6. Exploratory Analysis	Aggregate records by day, month, hour, weekday, and distribution company.
7. Modeling	Train and evaluate a Random Forest Regressor using date-time features.
8. Reporting	Generate charts, tables, insights, and export the final report as DOCX.

Machine Learning Design

The prediction notebook uses Year, Month, Day, DayOfWeek, and Hour as input features, with Total_Consumption as the target variable. The dataset is split into 80 percent training records and 20 percent testing records using random_state=42. A Random Forest Regressor with 100 estimators is used because it can model nonlinear relationships between time features and electricity consumption without requiring strong linearity assumptions.

Evaluation Measures

- Mean Absolute Error (MAE) measures average absolute prediction error.
- Root Mean Squared Error (RMSE) gives higher penalty to larger errors.
- R2 score indicates how much variation in consumption is explained by the model.

Exploratory Data Analysis

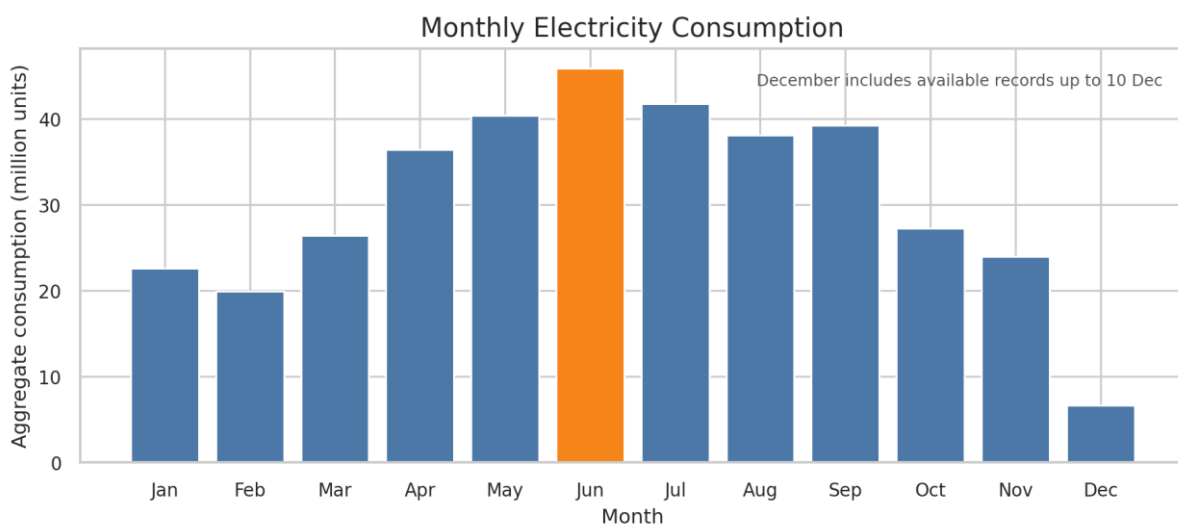


Figure 1: Monthly aggregate electricity consumption for available 2022 records.

Month	Total (million)	Share	Remark
January	22.63	6.14%	
February	19.91	5.40%	
March	26.40	7.16%	
April	36.47	9.89%	
May	40.42	10.96%	
June	45.94	12.46%	Highest
July	41.75	11.32%	
August	38.13	10.34%	
September	39.25	10.64%	
October	27.27	7.40%	
November	23.96	6.50%	
December	6.62	1.79%	Lowest / partial records

The highest monthly consumption is observed in June with 45.94 million aggregate units. December appears lowest because the available local files for December end on 10 December 2022.

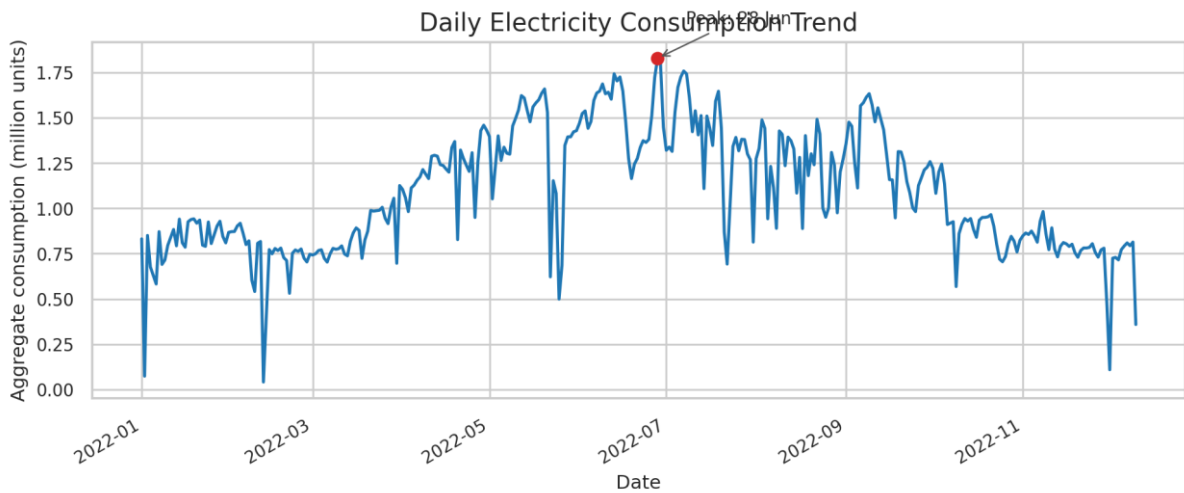


Figure 2: Daily trend showing variation and the maximum daily load period.

The highest daily aggregate consumption is 1,830,274.24 on 2022-06-28. The lowest daily aggregate consumption is 40,407.20 on 2022-02-12, which reflects the effect of incomplete or very limited available intervals in some daily files.

Temporal and Distribution Analysis

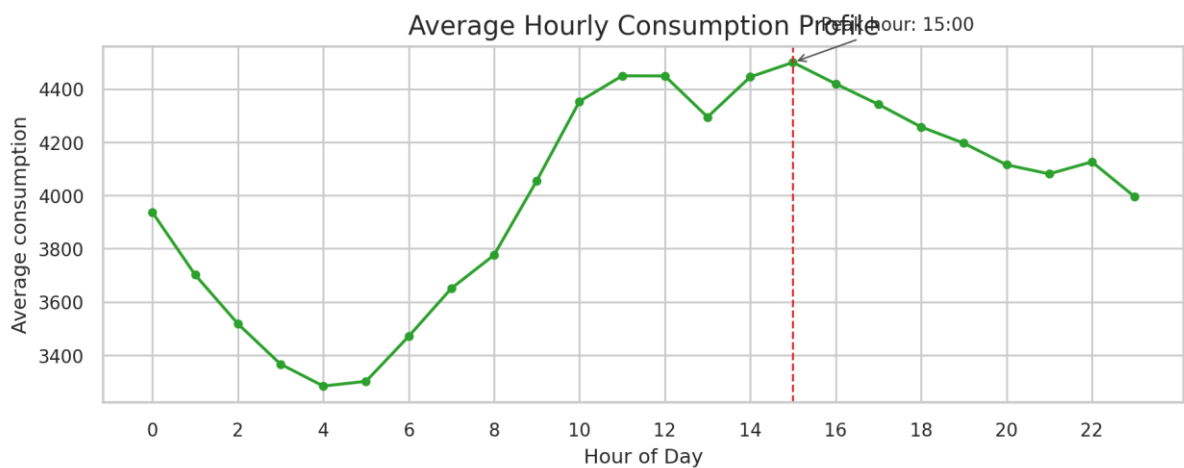


Figure 3: Average consumption by hour of day.

The average hourly profile indicates that demand rises through the day and reaches its highest average value around 15:00, with an average consumption of 4,499.55. The lowest average period is around 04:00.

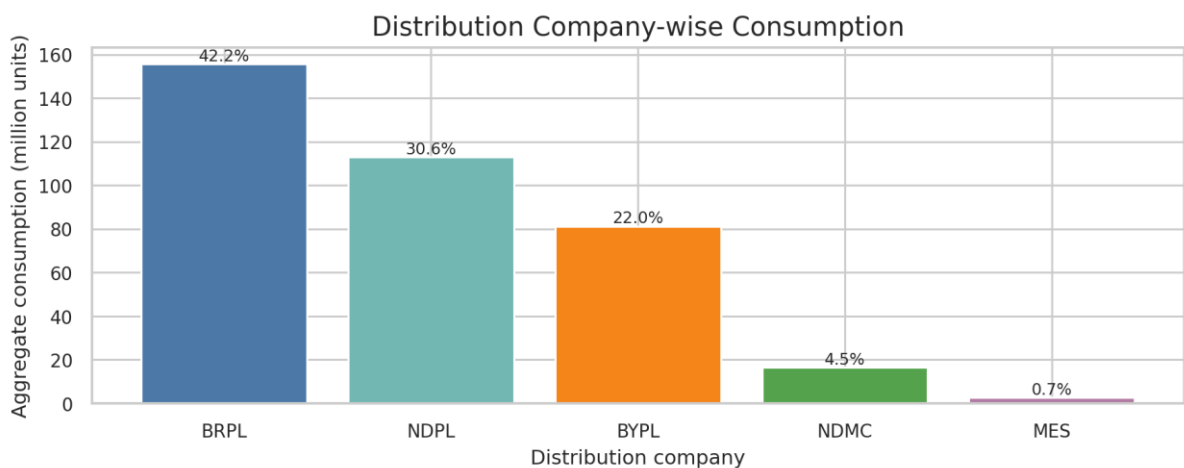


Figure 4: Distribution company-wise aggregate consumption.

Company	Total (million)	Share
BRPL	155.54	42.18%
NDPL	112.97	30.63%
BYPL	81.21	22.02%
NDMC	16.48	4.47%
MES	2.56	0.69%

BRPL contributes the largest share of aggregate consumption at 42.18 percent, followed by NDPL and BYPL. This indicates that distribution-area demand is uneven and should be considered in load planning.

Peak Load and Prediction Results

Date	Timeslot	Delhi	Total Consumption
2022-06-29	15:00	7,654.00	7,697.17
2022-06-29	15:05	7,673.23	7,600.28
2022-06-29	15:20	7,653.59	7,576.92
2022-06-29	15:15	7,647.11	7,575.88
2022-06-29	15:35	7,643.90	7,571.20

The top peak records are concentrated in the afternoon, which is consistent with the hourly profile. The correlation analysis also shows a near-perfect relationship between the DELHI reading and computed Total_Consumption (correlation = 1.0000).

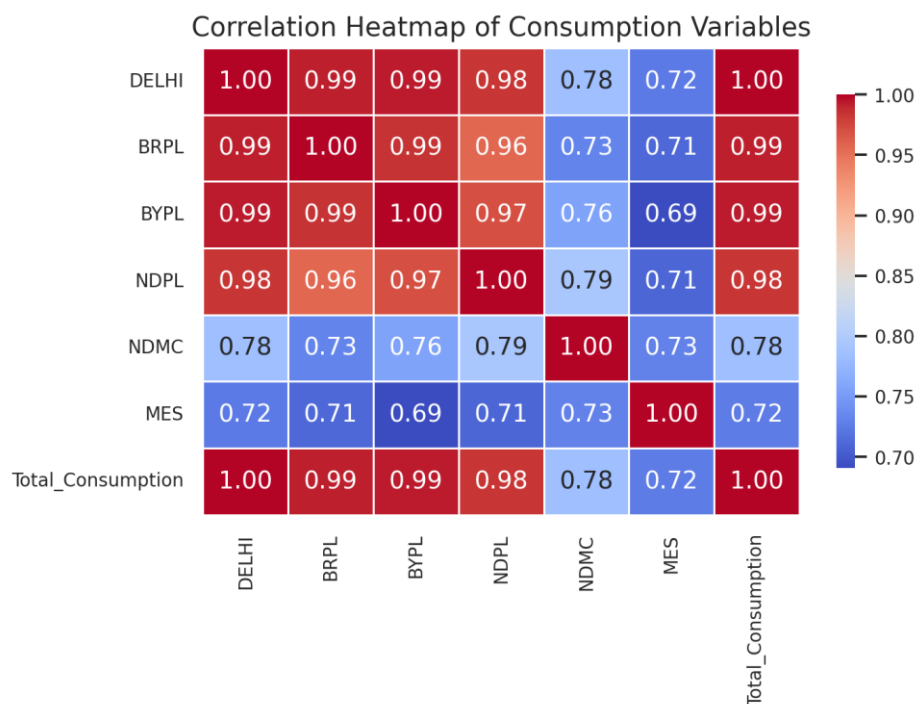


Figure 5: Correlation between Delhi, distribution company, and total variables.

Model Item	Result
Algorithm	Random Forest Regressor
Features	Year, Month, Day, DayOfWeek, Hour
Target	Total_Consumption
Training records	73,814

Testing records	18,454
MAE	59.89
RMSE	86.54
R2 Score	0.995306
Sample prediction	4,728.89

The Random Forest model achieved an R2 score of 0.995306, indicating that the selected date-time features explain most of the variation in the test data. The RMSE is 86.54, which is low relative to the observed peak values.

Model Visualization and Conclusion

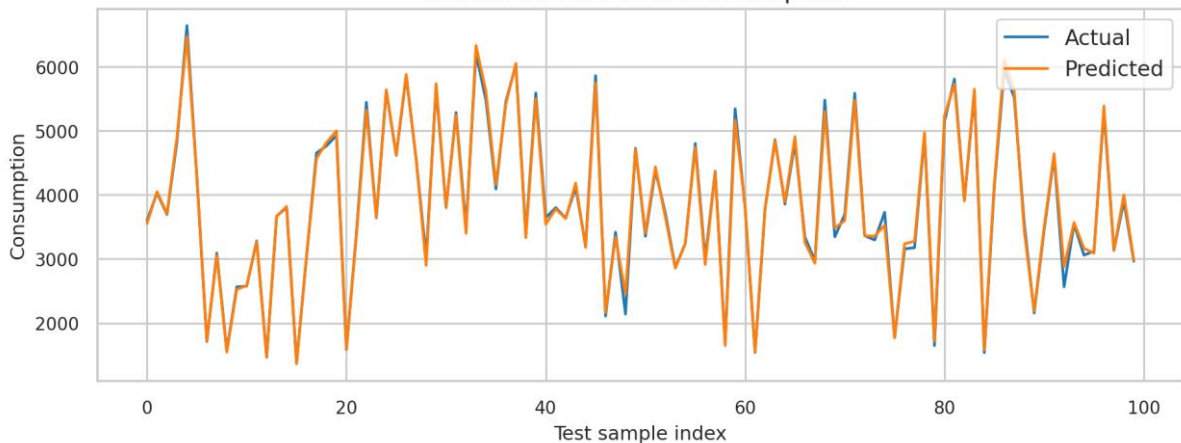


Figure 6: Actual vs predicted consumption for the first 100 test records.

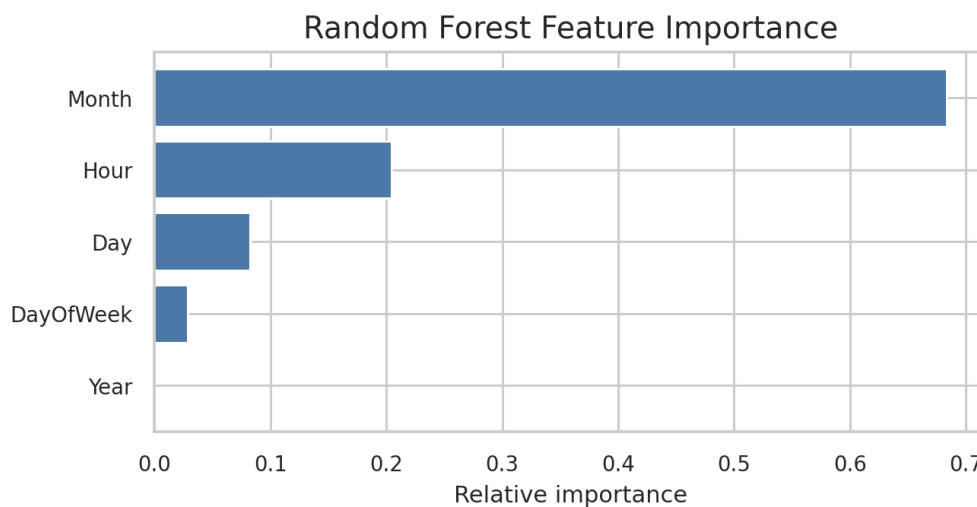


Figure 7: Random Forest feature importance.

Feature importance shows that Month and Hour are the strongest predictors. This supports the exploratory findings that electricity demand has both seasonal and intraday structure. For the sample input (Year=2022, Month=7, Day=1, DayOfWeek=4, Hour=18), the model predicts 4,728.89 consumption units.

Conclusion

- June is the highest-consumption month among the available local records, with 45.94 million aggregate units.
- The peak average hour is 15:00, while early morning hours show the lowest average consumption.

- BRPL is the largest distribution contributor, followed by NDPL and BYPL.
- The Random Forest Regressor performs strongly on the test set with an R2 score of 0.995306.
- The analysis confirms that time-based patterns can be used effectively for short-term electricity consumption prediction.

Limitations and Future Scope

- Some daily files contain fewer than the expected 288 five-minute intervals, so analysis is based on available records.
- The local dataset ends on 10 December 2022; future work can add the remaining December records if available.
- Prediction accuracy can be improved by adding weather, holiday, population, and industrial activity features.
- Advanced time-series models such as SARIMA, Prophet, LSTM, or gradient boosting can be compared with the Random Forest baseline.

References

Reference	Details
Kaggle Dataset	https://www.kaggle.com/datasets/adityasartape/12-months-data-of-delhi-electricity-board-2022/data?select=delhi_dispatch
Notebook/data_cleaning.ipynb	Data merging, date extraction, cleaning, and feature engineering.
Notebook/data_analysis.ipynb	Daily, weekly, monthly, hourly, company-wise, and correlation analysis.
Notebook/prediction_model.ipynb	Random Forest model training, evaluation, and prediction workflow.